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MEDGAR EVERS COLLEGE
The City University of New York

CS281 - 001: Database Systems

Quiz 3

No electronic devices or notes allowed.

Question 1: (20 points)

Write a query using NATURAL JOIN that produces the following result:

ID	name	dept_name	tot_cred	course_id	sec_id	semester	year	grade
00128	Zhang	Comp. Sci.	102	CS-101	1	Fall	2017	A
00128	Zhang	Comp. Sci.	102	CS-347	1	Fall	2017	A-
12345	Shankar	Comp. Sci.	32	CS-190	2	Spring	2017	A
12345	Shankar	Comp. Sci.	32	CS-315	1	Spring	2018	A
12345	Shankar	Comp. Sci.	32	CS-347	1	Fall	2017	A
54321	Williams	Comp. Sci.	54	CS-101	1	Fall	2017	A-
55739	Sanchez	Music	38	MU-199	1	Spring	2018	A-
76543	Brown	Comp. Sci.	58	CS-101	1	Fall	2017	A
76543	Brown	Comp. Sci.	58	CS-319	2	Spring	2018	A
98988	Tanaka	Biology	120	BIO-101	1	Summer	2017	A

```
Select ,ID, name, dept_name, tot_cred , course_id, sec_id, Semester,
year, grade
from takes natural join student
Where grade Like 'A%';
```

NAME Akenaya Chuoh

Question 2: (20 points)

Rewrite the following query with JOIN … USING instead of NATURAL JOIN:

```
SELECT name, id, transcript, address_number
FROM person
NATURAL JOIN (SELECT person_id AS id, transcript
              FROM interview)
WHERE address_street_name = 'Northwestern Dr'
AND address_number = (SELECT MAX(address_number)
                      FROM person
                      WHERE address_street_name = 'Northwestern Dr');
```

```
Select name, id, transcript, address-number, person_id AS id
from person
  Join interview ON person.id = interview.id
  where address_street_name = 'Northwestern Dr'
  AND address_number = (select Max (address_number)
                        from person
                        where address_street_name = 'Northwestern
                        Dr';
```

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Question 3: (20 points)

Rewrite the following query with JOIN … ON instead of NATURAL JOIN:

```
SELECT *
FROM course NATURAL JOIN section
WHERE course_id LIKE '%-101'
OR title LIKE 'Intro.%';
```

```
Select *
from course
Join Section
  ON    Section.course_id = course.course_id

where course_id Like `%101'
  or title Like `Intro%';
```

Question 4: (20 points)

Rewrite the following query with JOIN … USING instead of NATURAL JOIN:

```
SELECT *
FROM student
NATURAL JOIN takes
NATURAL JOIN (SELECT course_id, title FROM course);
```

```
Select *
from Student
Join takes
  ON student.ID = takes.ID
```

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Question 5: (20 points)

Consider the following employee database:

employee (id, person_name, street, city, state)
company (company_name, city, state)
works (id, company_name, salary)

Write create table statements for the schema provided above. The underlined attributes are primary keys.

- Provide primary keys and foreign keys.
- Only salary should be numeric. All other attributes should be varchar.
- All attributes should reject null values. (Primary keys reject null values by default.)
- Ensure the id attribute is exactly 7 characters longs.
- Ensure state is exactly 2 characters long.
- Ensure salary is no less than $40,000 and no greater than $200,000.

```
Create table employee (
(re  id  varchar (7)
   person-name  varchar (20) not null
   street  varchar (17) not null
   City  varchar (25) notnull
   state  varchar (2) notnull
   primary key (id)
   foreign key (id, state) referenced from Company
)

create table works (
   id  varchar (7)
   Company-name varchar (50)
   salary numeric (6, 0)
   salary > 40000 AND
   salary < 200000
   primary key (id)
   ; foreignkey (company-name)
   fron Company

create table Company (
   Company-name  var char (50)
   City  varcher (25)
   -state  varcher (2)
   primary key (Company-name, city, state) ;
)
```

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Extra Credit: (10 points)

Rewrite the following query. You must show 2 queries. One with NATURAL JOIN. The other with USING. Attributes for USING should be identical to those of NATURAL JOIN.

```
SELECT student.ID AS ID, student.name, student.dept_name , tot_cred, instructor.ID AS instructor_id,
        instructor.name AS instructor_name, instructor.dept_name AS instructor_dept_name, salary
FROM student
JOIN advisor
ON student.ID = s_id
JOIN instructor
ON i_id = instructor.ID
WHERE tot_cred >= 80;
```